

Supplemental statistical tables

Doc-friendly version for copying into a manuscript document. Tables are generated from the CSV files in outputs/publication_figures/supplemental_tables/.

S Table 1. Survey analysis input summary

Summary of survey rows, sites, effort, observations, and length measurements used in the statistical analyses.

Quantity	Value
Survey rows	3,978
Survey sites	133
Basins	9
Survey-level habitat categories	6
Total survey effort, hours	848.0
Total <i>P. helianthoides</i> observed	2,651
Rows with zero <i>P. helianthoides</i>	85.3%
Positive individual length measurements	1,954
Sites with positive length measurements	48

S Table 2. Site-level eelgrass association across count models

Incidence-rate ratios compare surveys at sites where eelgrass was recorded with surveys at sites where eelgrass was not recorded. All models include a survey-effort offset.

Model	Family	Adjustment	IRR for eelgrass-at-site	95% CI	p	Rows	Sites
NB: eelgrass only	Negative binomial	Survey effort only	8.78	3.59–21.46	<0.001	3,978	133
NB: + habitat	Negative binomial	Survey effort + habitat	5.15	2.45–10.84	<0.001	3,978	133
NB: + basin	Negative binomial	Survey effort + basin	6.06	1.67–21.95	0.006	3,978	133
NB: + basin + habitat	Negative binomial	Survey effort + basin + habitat	4.28	1.49–12.27	0.007	3,978	133
Poisson: eelgrass only	Poisson	Survey effort only	9.03	3.80–21.47	<0.001	3,978	133
Poisson: + habitat	Poisson	Survey effort + habitat	5.57	2.65–11.69	<0.001	3,978	133

Poisson: + basin	Poisson	Survey effort + basin	6.32	2.23–17.90	<0.001	3,978	133
Poisson: + basin + habitat	Poisson	Survey effort + basin + habitat	3.84	1.65–8.94	0.002	3,978	133

S Table 3. Primary adjusted count-model terms

Primary negative-binomial model adjusted for basin and survey-level habitat. Reference categories were Central basin and Natural Reef habitat.

Primary adjusted model term	IRR	95% CI	p
Basin: Admiralty Inlet	4.13	1.15–14.89	0.030
Basin: Hood Canal	36.39	9.13–145.14	<0.001
Basin: Howe Sound	2.07	0.37–11.55	0.406
Basin: San Juan	1.25	0.24–6.45	0.787
Basin: South	0.21	0.04–1.14	0.070
Basin: Strait of Georgia	18.00	5.56–58.30	<0.001
Basin: Strait of Juan de Fuca	18.92	7.24–49.43	<0.001
Basin: Whidbey	27.96	10.31–75.85	<0.001
Habitat: Artificial Reef	1.17	0.41–3.33	0.769
Habitat: Eelgrass	1.34	0.63–2.87	0.450
Habitat: Kelp Forest	0.64	0.29–1.43	0.281
Habitat: Soft Bottom	3.12	1.31–7.42	0.010
Habitat: Sponge Garden	0.00	0.00–0.00	<0.001
Eelgrass recorded at site	4.28	1.49–12.27	0.007

S Table 4. Descriptive habitat-by-eelgrass and basin-by-eelgrass cells

*Sparse descriptive cells had fewer than 3 sites, fewer than 20 survey rows, or zero total *P. helianthoides*. Sparse flags are descriptive subgroup cautions, not exclusions from the count models.*

Grouping	Group	Eelgrass status	Rows	Sites	Survey hours	<i>P. helianthoides</i>	Mean encounter rate hr ⁻¹	SE	Sparse descriptive cell
Habitat × eelgrass	Soft Bottom	Eelgrass at site	536	32	105.0	1,484	12.97	1.05	No
Habitat × eelgrass	Soft Bottom	No eelgrass at site	316	41	52.0	73	1.39	0.37	No
Habitat × eelgrass	Eelgrass	Eelgrass at site	243	34	31.2	175	4.10	0.67	No

Habitat × eelgrass	Artificial Reef	Eelgrass at site	548	16	131.5	523	3.14	0.44	No
Habitat × eelgrass	Artificial Reef	No eelgrass at site	468	17	126.4	39	0.20	0.06	No
Habitat × eelgrass	Kelp Forest	Eelgrass at site	48	4	13.2	31	1.93	0.57	No
Habitat × eelgrass	Kelp Forest	No eelgrass at site	59	17	13.6	2	0.03	0.03	No
Habitat × eelgrass	Natural Reef	Eelgrass at site	368	20	80.1	95	1.30	0.28	No
Habitat × eelgrass	Natural Reef	No eelgrass at site	1,377	78	290.9	229	0.77	0.13	No
Habitat × eelgrass	Sponge Garden	Eelgrass at site	3	1	1.0	0	0.00	0.00	Yes
Habitat × eelgrass	Sponge Garden	No eelgrass at site	12	2	3.1	0	0.00	0.00	Yes
Basin × eelgrass	Howe Sound	Eelgrass at site	21	1	3.8	0	0.00	0.00	Yes
Basin × eelgrass	Howe Sound	No eelgrass at site	158	9	30.0	4	0.20	0.14	No
Basin × eelgrass	Strait of Georgia	Eelgrass at site	4	1	0.5	3	5.73	3.31	Yes
Basin × eelgrass	Strait of Georgia	No eelgrass at site	4	1	0.5	0	0.00	0.00	Yes
Basin × eelgrass	San Juan	Eelgrass at site	142	2	23.2	0	0.00	0.00	Yes
Basin × eelgrass	San Juan	No eelgrass at site	656	35	121.5	14	0.08	0.03	No
Basin × eelgrass	Strait of Juan de Fuca	Eelgrass at site	177	4	40.3	200	4.36	0.78	No
Basin × eelgrass	Strait of Juan de Fuca	No eelgrass at site	18	5	10.1	20	4.42	2.24	Yes
Basin ×	Admiralty	Eelgrass at	90	3	15.9	7	0.57	0.28	No

eelgrass	Inlet	site							
Basin × eelgrass	Admiralty Inlet	No eelgrass at site	270	5	85.6	33	0.53	0.27	No
Basin × eelgrass	Whidbey	Eelgrass at site	506	6	122.5	1,940	16.88	1.13	No
Basin × eelgrass	Whidbey	No eelgrass at site	363	6	87.9	6	0.06	0.03	No
Basin × eelgrass	Central	Eelgrass at site	501	10	98.3	15	0.20	0.10	No
Basin × eelgrass	Central	No eelgrass at site	350	15	73.5	24	0.18	0.09	No
Basin × eelgrass	South	Eelgrass at site	162	3	29.5	2	0.07	0.05	No
Basin × eelgrass	South	No eelgrass at site	59	4	10.5	0	0.00	0.00	Yes
Basin × eelgrass	Hood Canal	Eelgrass at site	143	4	27.9	141	5.17	0.77	No
Basin × eelgrass	Hood Canal	No eelgrass at site	354	19	66.5	242	3.39	0.51	No

S Table 5. Length distribution-shape analyses

KS D is the two-sample Kolmogorov-Smirnov distance. Energy distance is a complementary distribution-shape statistic. Overall and habitat-specific site-eelgrass tests use site-label permutation p-values; S2 within-panel habitat-pair rows show site-cluster bootstrap intervals and individual-level KS q-values only as sensitivity values.

Panel/ contrast	Comparison	N individuals	N sites	KS D	KS CI or p/q	Energy distance	Energy CI or p/q	Sparse
Soft Bottom only	Eelgrass at site vs No eelgrass at site	1,052 vs 59	11 vs 10	0.623	p=0.001; q=0.006	4.009	p=0.001; q=0.005	No
Natural Reef only	Eelgrass at site vs No eelgrass at site	98 vs 180	9 vs 23	0.316	p=0.370; q=0.616	1.953	p=0.344; q=0.573	No
Overall	Eelgrass at	1,698 vs 256	17 vs 31	0.259	p=0.110;	1.600	p=0.051;	No

positive length distribution	site vs No eelgrass at site				q=0.275		q=0.126	
Kelp Forest only	Eelgrass at site vs No eelgrass at site	22 vs 2	2 vs 1	0.409	p=1.000; q=1.000	1.828	p=1.000; q=1.000	Yes
Artificial Reef only	Eelgrass at site vs No eelgrass at site	384 vs 15	5 vs 5	0.353	p=0.659; q=0.824	1.566	p=0.643; q=0.803	Yes
S2 panel: Eelgrass at site	Eelgrass vs Artificial Reef	142 vs 384	6 vs 5	0.513	0.22–0.81	2.475	0.85–4.77	No
S2 panel: Eelgrass at site	Soft Bottom vs Eelgrass	1,052 vs 142	11 vs 6	0.459	0.20–0.59	2.038	0.80–2.72	No
S2 panel: Eelgrass at site	Eelgrass vs Natural Reef	142 vs 98	6 vs 9	0.441	0.21–0.91	3.112	1.04–6.21	No
S2 panel: Eelgrass at site	Soft Bottom vs Natural Reef	1,052 vs 98	11 vs 9	0.406	0.24–0.86	2.192	1.26–5.12	No
S2 panel: Eelgrass at site	Artificial Reef vs Natural Reef	384 vs 98	5 vs 9	0.371	0.23–0.81	1.970	1.20–4.69	No
S2 panel: Eelgrass at site	Soft Bottom vs Artificial Reef	1,052 vs 384	11 vs 5	0.123	0.05–0.67	0.549	0.17–3.44	No
S2 panel: No eelgrass at site	Soft Bottom vs Natural Reef	59 vs 180	10 vs 23	0.502	0.18–0.72	3.820	0.89–5.37	No

S Table 6A. ShoreZone model performance

Performance summary for the exploratory ShoreZone random-forest model.

Metric	Value
Survey sites with complete ShoreZone features	104
One-hot encoded ShoreZone features	77
Cross-validated R ² , mean ± SD	0.104 ± 0.263
Prediction vs observed Pearson r	0.291
Prediction vs observed p	0.003
RMSE, individuals hr ⁻¹	5.08
MAE, individuals hr ⁻¹	2.64

S Table 6B. Top ShoreZone prediction-loss features

Features with the largest mean prediction loss when permuted.

Feature	Prediction loss	SE
Eelgrass: continuous	0.292	0.065
BC shoreline type: Sand beach	0.069	0.028
Low-exposure sand/gravel habitat	0.050	0.013
Oyster beds: absent	0.036	0.006
Oyster beds: patchy	0.009	0.004
Fucus band: patchy	0.009	0.002
Eelgrass: absent	0.005	0.002
Supratidal + two intertidal + subtidal components	0.005	0.003
Eelgrass: patchy	0.004	0.003
Ulva green algae: absent	0.003	0.002

S Table 7A. Protected-exposure by continuous-eelgrass group summaries

Site-level encounter-rate summaries by ShoreZone exposure group and continuous mapped eelgrass status.

Exposure group	ShoreZone eelgrass group	Sites	Mean encounter rate hr ⁻¹	SE	Median encounter rate hr ⁻¹	P. helianthoides	Survey hours
Other exposure	Eelgrass: continuous	2	1.00	1.00	1.00	1	1.7
Other exposure	Other	62	1.01	0.32	0.00	336	429.6
Protected / very protected	Eelgrass: continuous	6	15.29	5.97	11.75	1,931	110.6

Protected / very protected	Other	41	1.80	0.45	0.10	333	269.6
----------------------------	-------	----	------	------	------	-----	-------

S Table 7B. Protected-exposure and continuous-eelgrass group tests

Non-parametric tests comparing ShoreZone exposure and continuous eelgrass groups.

Contrast	Test	Statistic	p
protected_or_very_protected_vs_other_exposure	Mann-Whitney U	1863.00	0.023
continuous_eelgrass_vs_other_zos	Mann-Whitney U	631.00	0.008
protected_continuous_eelgrass_vs_all_other	Mann-Whitney U	532.50	0.003
four_fig4b_groups	Kruskal-Wallis	11.42	0.010

S Table 7C. ShoreZone eelgrass versus survey-recorded eelgrass agreement tests

Agreement tests comparing ShoreZone eelgrass classes with survey-recorded site-level eelgrass status.

Agreement contrast	Test	Statistic	df	p	Effect	Effect value
shorezone_three_category_by_diver_binary	chi-square	12.09	2	0.002	Cramers_V	0.341
shorezone_any_eelgrass_by_diver_binary	Fisher exact	3.55		0.005	phi	0.265